Michael Jay

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EDUCATION

University of California, Los Angeles *June 2028*
Mechanical Engineering B.S. GPA: 3.83 / 4.0
Relevant Coursework: Mechanisms and Mechanical Systems, Advanced Strength of Materials, Dynamics of Rigid Bodies

SKILLS

- **CAD/FEA:** SolidWorks (CSWA Certified, FEA), Autodesk Fusion 360, ANSYS
- **Technical:** CNC Machining, Manual Lathe & Mill, Additive Manufacturing (FDM/SLA/Resin), GD&T, Soldering, GitHub, Arduino Uno
- **Software:** Chitubox, Bambu Studio, Python, Java, C++, MATLAB, Excel
- **Additional:** Spanish (Professional); Machine Shop Safety & Equipment Maintenance Training

EXPERIENCE

ESP32 Drone With Custom Flight Controller **Palm Desert, CA**
Lead Designer & Engineer *May 2026 – Present*
- Recreated and modified a custom drone frame inside SolidWorks, applying GD&T principles to arm mounts, structural housing, and fastener clearances to ensure precise component fitment during physical assembly.
- Wired and soldered an ESP32 microcontroller to a 4-in-1 ESC, MPU6050 accelerometer, and buck converter, writing C++ firmware to process I2C motion sensor data and convert input signals into responsive motor outputs.
- Fabricated custom frame prototypes using FDM 3D printing, iterating the arm geometry and evaluating higher temperature material options after observing thermal deformation from extreme heat exposure during field testing.

Baja SAE at UCLA **Los Angeles, CA**
Chassis Project Engineer *August 2025 – Present*
- Engineered custom side panels, footbox, and SIMs firewall in SolidWorks, mating components to the top chassis assembly and strategically repositioning tab locations to optimize structural rigidity, driver safety, and serviceability.
- Secured $5,000+ from Nexx Technologies for carbon fiber prepreg material donations by pitching regional industry suppliers and managing corporate partner relationships to minimize subteam manufacturing costs.
- Cross-collaborated with subteam leads and system integration groups to resolve packaging interference, streamline panel mounting workflows, and adhere to strict SAE design compliance standardizations.

Creative Labs at UCLA **Los Angeles, CA**
Project Lead - Reactive Subway Map *September 2026 – Present*
- Designed precision laser-cut acrylic transit routes on a custom 3D-printed baseboard, applying DFM guidelines to streamline part manufacturing, optimize component alignment, and maintain clean internal wire management.
- Programmed an ESP32 microcontroller in C++ to fetch real-time LA Metro public transit API data over Wi-Fi, converting live movement feeds into dynamic LED signal updates across addressable lighting strips.
- Recruited and led a multidisciplinary team of underclassmen through hardware assembly and programming, actively mentoring members on digital fabrication equipment, 3D printing workflows, and parametric CAD design techniques.

Westwood SolidWorks User Group **Los Angeles, CA**
Workshop Leader *January 2026 - Present*
- Lead weekly workshops to help students master SolidWorks, breaking down complex modeling techniques into clear, structured lessons, and presenting live design demonstrations to groups of 20+ members, managing technical Q&A and guiding peers through live troubleshooting of complex design errors.
- Collaborated with fellow organizers to shape the curriculum, ensuring each session provides a project that effectively builds member proficiency in professional-grade mechanical design and parametric modeling.

RESEARCH

DiCarlo Lab at California NanoSystems Institute **Los Angeles, CA**
Undergraduate Researcher *January 2026 – Present*
- Optimizing an automated decapping system for arrays of biomedical samples, influencing mechanical design iterations to improve accuracy and operational speed and torque consistency.
- Leveraging Autodesk Fusion 360 to model intricate assemblies and refine specific components for improved integration into the full device, while maintaining high-precision clearances.
- Fabricating high-fidelity prototypes using FDM and SLA (resin) 3D printing to consider mechanical tolerances and iterate on part functionality, for high-throughput laboratory environments.